

energy per 4 ms bin (nJ)
cell delays annotated, 5,000-bin window

- H = 16
- H = 32
- H = 48
- H = 64
- H = 128
- constants
- latches
- SRAM block
- software
- equal H, density varied
- dense, H varied

comparison
threshold 0.55

NeuroBench SNN2
(float) 0.593

software, hand-tuned
(100-bin decoding window)

SRAM, dense mode

SRAM, event mode

SRAM, gated groups

64, 100 %, 16-bit
64, 100 %
64, 100 %, 16-bit gated
48, 100 %
latch memory
32, 100 %
48, 50 %
64, 25 %
128, 100 %
64, 50 %
48, 25 %
32, 25 %
16, 100 %
16, 50 %

128, 12.5 %

64, 12.5 %

32, 12.5 %

test R^2 , three-session mean (integer reference)

0.48 0.50 0.52 0.54 0.56 0.58 0.60 0.62